CNDO/2 Based Calculation of Polarity and Polarizability for Molecular Classification

CHEM 279 - Final Presentation

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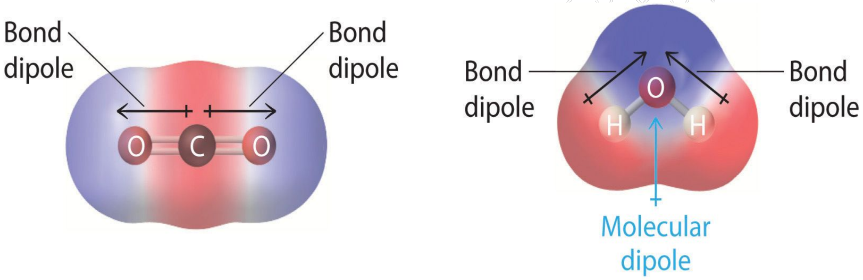
Objectives

- Develop molecular classification pipeline derived from CNDO/2 theory

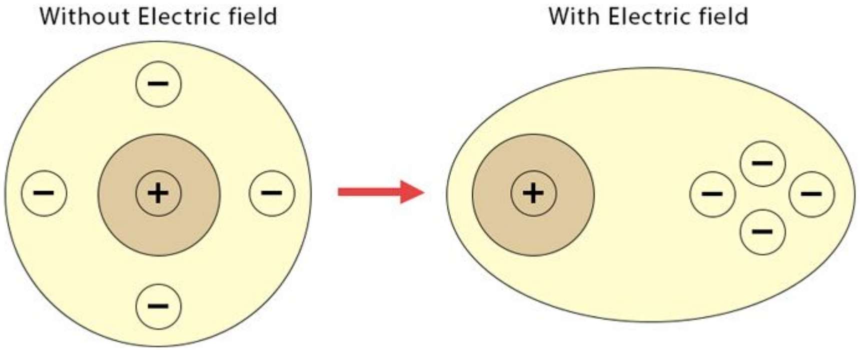
- Calculate dipole vector and polarizability tensor

- Extract molecular properties
  - Permanent Dipole Magnitude
  - Isotropic Polarizability
  - Anisotropic Polarizability



(a) No net dipole moment

(b) Net dipole moment



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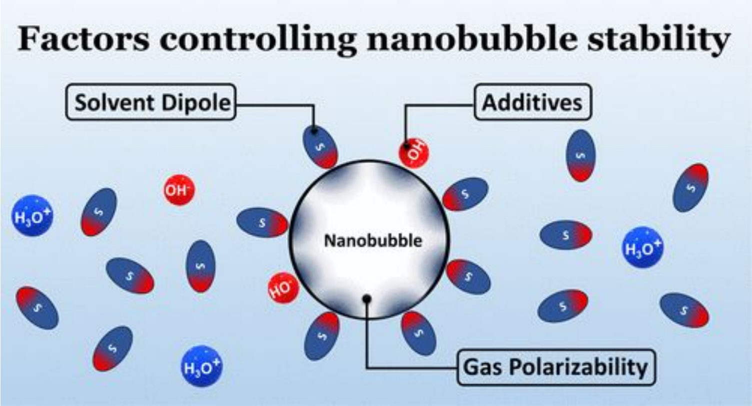
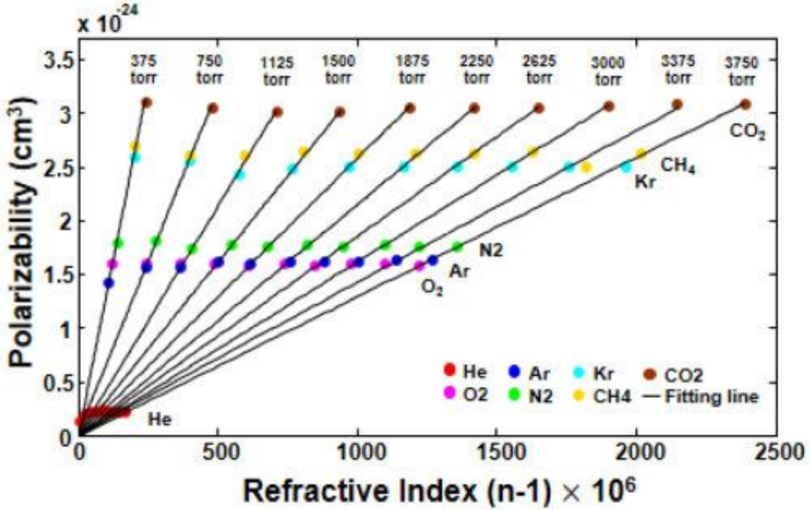
Motivation

- Permanent Dipole Magnitude
  - Solvent Behavior
  - Dielectric constant
  - Spectroscopy
- Isotropic Polarizability
  - Refractive Index
  - Optical response
- Anisotropic Polarizability
  - Polarized Light
  - Molecular alignment in fields





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Dipole Theory

Theory

Dipole (μ) is first derivative of energy with respect to external electric field (F)

$$\mu = -\frac{\partial E}{\partial F} \qquad (1)$$

General form of energy derivative

$$\frac{\partial E}{\partial F} = \underbrace{\langle \psi \mid \frac{\partial \hat{H}}{\partial F} \mid \psi \rangle}_{\text{Electronic term}} + \underbrace{\frac{\partial E_{nuc}}{\partial F}}_{\text{Nuclear term}} \qquad (2)$$

Hamiltonian as a function of F

$$\hat{H}(F) = \hat{H}_0 + F_x \hat{x} + F_y \hat{y} + F_z \hat{z} \qquad (3)$$

Hamiltonian derivative

$$\frac{\partial \hat{H}}{\partial F_x} = \hat{x} \qquad (4)$$

Substitute into energy derivative

$$\frac{\partial E}{\partial F_x} = \langle \psi \mid \hat{x} \mid \psi \rangle + \frac{\partial E_{nuc}}{\partial F_x} \qquad (5)$$

CNDO/2 Representation

Full dipole equation

$$\mu_x = -\frac{\partial E}{\partial F_x} = -\left(\underbrace{\sum_{\mu\nu} P_{\mu\nu} D^x_{\mu\nu}}_{\text{Electronic term}} + \underbrace{\sum_A Z_A R_{A,x}}_{\text{Nuclear term}} \right) \qquad (6)$$

Dipole matrix formula

$$D^x_{\mu\nu} = \langle \chi_\mu \mid x \mid \chi_\nu \rangle \qquad (7)$$

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Dipole Integral (D$\mu\nu$)

Dipole matrix formula,
AO basis representation

$$D^x_{\mu\nu} = \langle \chi_\mu \mid x \mid \chi_\nu \rangle \quad (1)$$

Overlap matrix formula

$$S_{\mu\nu} = \langle \chi_\mu \mid \chi_\nu \rangle \quad (2)$$

Dipole matrix in terms
of STO3G Basis

$$D^x_{\mu\nu} = \sum_k \sum_l d_{k\mu} d_{l\nu} N_{k\mu} N_{l\nu} D^{kl}_x \quad (3)$$

Primitive Gaussian A

$$g_A = (x - X_A)^{l_A} (y - Y_A)^{m_A} (z - Z_A)^{n_A} e^{\alpha \mid r - R_A \mid^2} \quad (4)$$

Primitive Gaussian B

$$g_B = (x - X_B)^{l_B} (y - Y_B)^{m_B} (z - Z_B)^{n_B} e^{\beta \mid r - R_B \mid^2} \quad (5)$$

Dipole Integral

$$D^{AB}_x = \int g_A(r) \cdot x \cdot g_B(r) dr \quad (6)$$

Formula for dipole terms

$$D^{AB}_x = [S^{AB}_x(l_A + 1, l_B) + X_A S^{AB}_x(l_A, l_B)] S^{AB}_y S^{AB}_z \quad (7)$$

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Polarizability Tensor Theory

Polarizability (α) is second derivative of energy with respect to external electric field (F)

$$\alpha = \frac{\partial^2 E}{\partial F^2}$$

Estimate using change in dipole with an applied electric field

$$\alpha_{ij} = \left(\frac{\partial \mu_i}{\partial F_j}\right)_{F=0}$$

Central differences approximation

$$\alpha_{ij} \approx \frac{\mu_i(+F_j) - \mu_i(-F_j)}{2F}$$

Process

Apply small +/– F perturbations

↓

Update fock and hamiltonian

$$\Delta h_{\mu\nu} = F_x D^x_{\mu\nu} + F_y D^y_{\mu\nu} + F_z D^z_{\mu\nu}$$

↓

Run SCF, compute dipole

↓

Calculate α_{ij} element

↓

Polarizability Tensor

$$\alpha = \begin{pmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{pmatrix}$$

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Derived Properties

$$\alpha = \begin{pmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{pmatrix}$$

Dipole Strength (μ)

- How polar the molecule is

$$\mu = |\mu|$$

Isotropic Polarizability ($\bar{\alpha}$)

- How easily molecule is polarized by external electric field

$$\bar{\alpha} = \frac{\alpha_{xx} + \alpha_{yy} + \alpha_{zz}}{3}$$

Anisotropic Polarizability ($\Delta\alpha$)

- How directional the response to the external electric field is

$$\Delta\alpha = \left[\frac{(\alpha_{xx} - \alpha_{yy})^2 + (\alpha_{yy} - \alpha_{zz})^2 + (\alpha_{zz} - \alpha_{xx})^2 + 6(\alpha_{xy}^2 + \alpha_{yz}^2 + \alpha_{xz}^2)}{2}\right]^{\frac{1}{2}}$$

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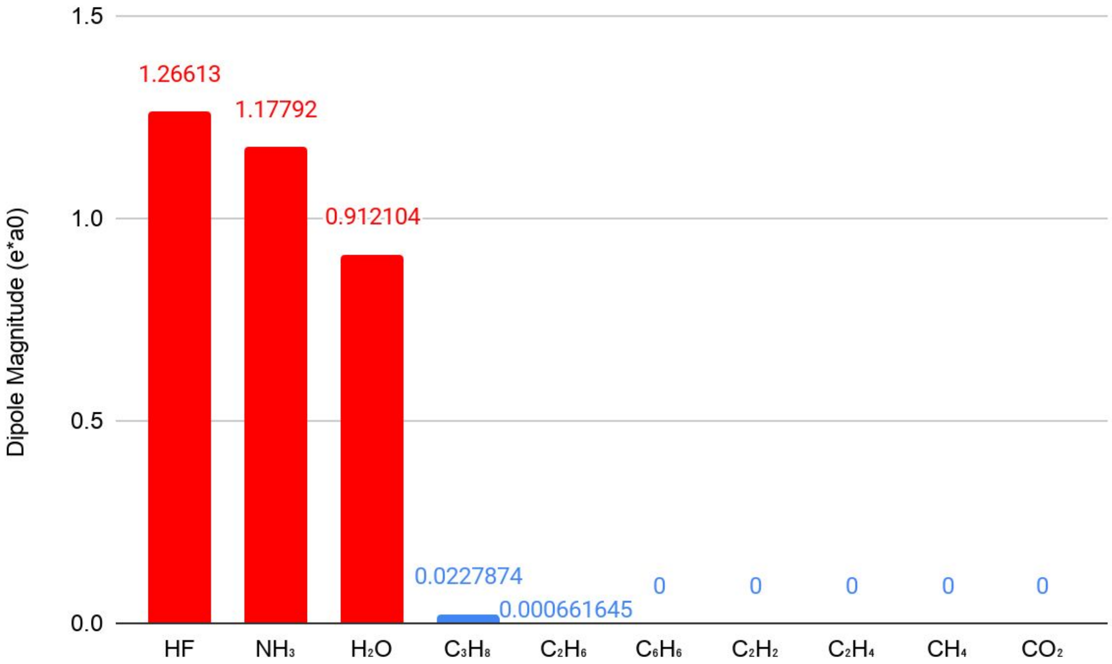
Molecules of Interest

Trends:

- Symmetry
  - Isotropic vs. Anisotropic
- Polarity
  - Geometry, lone pairs, electronegativity
- Size
  - Polarizability ↑ as Size ↑
- Delocalization of electrons
  - Polarizability ↑ as Number of π–bonds ↑

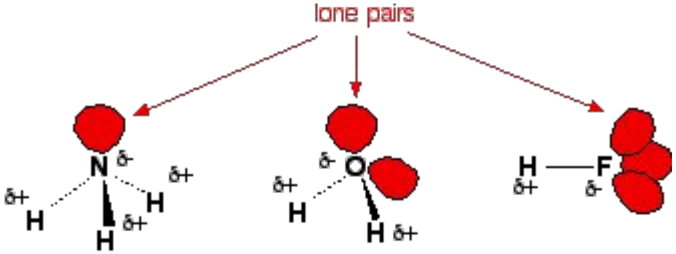
Name	Symbol	Symmetric?	Polarity	Number of π–bonds
Methane	CH_4	✓	Nonpolar	0
Ethane	C_2H_6	✓	Nonpolar	0
Propane	C_3H_8		Nonpolar	0
Acetylene	C_2H_2	✓	Nonpolar	2
Ethylene	C_2H_4	✓	Nonpolar	1
Benzene	C_6H_6	✓	Nonpolar	3
Carbon Dioxide	CO_2	✓	Nonpolar	2
Water	H_2O		Polar	0
Ammonia	NH_3		Polar	0
Hydrogen Fluoride	HF		Polar	0

Dipole Results



Attributes affecting |μ|

- Number of lone pairs
- Geometry
- Electronegativity



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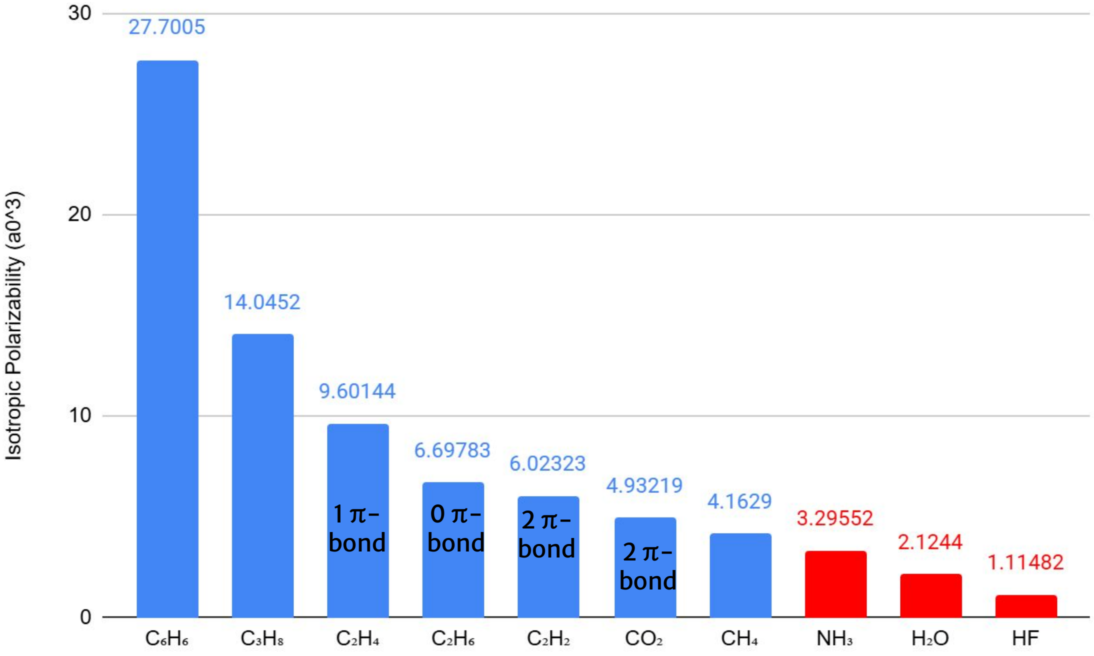
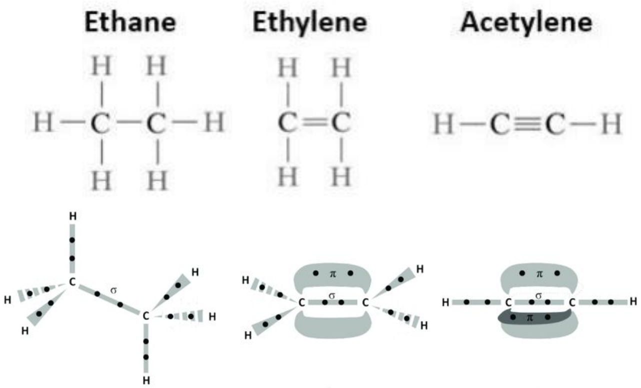
All data in au

Isotropic Polarizability Results



- Size dominates
- Acetylene is non-intuitive



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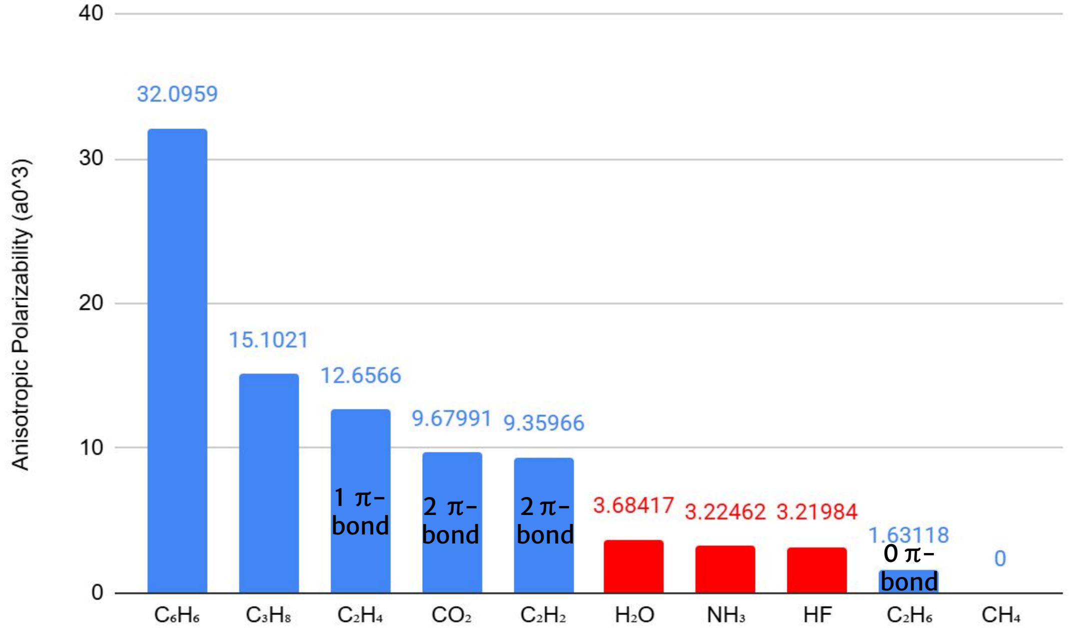
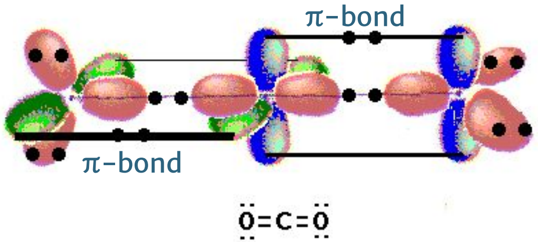
Anisotropic Polarizability Results



- Size dominates
- Symmetry factor
- CO_2
  - 9.68 (aniso) vs 4.93 (iso)



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Molecular Classification

Polar Candidates (↑μ)

1. HF
2. NH_3
3. H_2O

Dielectric Candidates (↑ᾱ, ↑μ)

1. C_6H_6
2. C_3H_8
3. NH_3

Optical Response Candidates (↑ᾱ)

1. C_6H_6
2. C_3H_8
3. C_2H_4

Directional Response Candidates (↑Δα)

1. C_6H_6
2. C_3H_8
3. C_2H_4

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Extensions for Final Report

- Vary strength of electric field
  - $|\mu|$, $\overline{\alpha}$, $\Delta\alpha$ should not depend on strength of external field, F

- Quantitative comparisons to literature values

- Polarizability Anisotropy Ratio= $\dfrac{\Delta\alpha}{\overline{\alpha}}$
  - Removes dependence on size of molecule
  - >1 : highly directional response
  - 0 : symmetric electron cloud

Thank You

Questions?

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